

Experiment No - 12

Simulate a) Analysis of ICMP and PING messages b) Analysis of ICMP and Trace route.

a) Analysis of ICMP and PING Messages

1. Overview of ICMP and PING

ICMP (Internet Control Message Protocol) is a core part of the Internet Protocol (IP) suite. It's used primarily for diagnostic and error-reporting purposes in network communications. One of the most common uses of ICMP is the **PING** utility, which helps determine whether a host on a network is reachable.

When you run the `ping` command on a device, it sends ICMP Echo Request messages to a destination IP address, and the destination replies with ICMP Echo Reply messages. The results give insight into network connectivity, round-trip time, and packet loss.

2. Example of ICMP Ping Messages

Let's simulate what happens when you ping a remote server (say, `google.com`):

- **Ping Command:** `ping google.com`

Here's a step-by-step breakdown of the ICMP messages:

- **Step 1:** The device (source) sends an **ICMP Echo Request** to `google.com`.
- **Step 2:** The destination server (`google.com`) receives the Echo Request, processes it, and then sends an **ICMP Echo Reply** back to the source device.
- **Step 3:** The source device receives the Echo Reply and records the round-trip time (RTT), which measures how long it took for the message to go from source to destination and back.

The typical output might look like this:

```
Pinging google.com [142.250.181.14] with 32 bytes of data:
Reply from 142.250.181.14: bytes=32 time=20ms TTL=54
Reply from 142.250.181.14: bytes=32 time=21ms TTL=54
Reply from 142.250.181.14: bytes=32 time=20ms TTL=54
Reply from 142.250.181.14: bytes=32 time=19ms TTL=54
```

```
Ping statistics for 142.250.181.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 19ms, Maximum = 21ms, Average = 20ms
```

3. Breakdown of PING Process

- **ICMP Echo Request:**
 - **Type:** 8 (Echo Request)
 - **Code:** 0
 - **Checksum:** Used for error checking.
 - **Identifier and Sequence Number:** Used to match requests and replies.
- **ICMP Echo Reply:**
 - **Type:** 0 (Echo Reply)
 - **Code:** 0
 - **Checksum:** Used for error checking.
 - **Identifier and Sequence Number:** These match the request to identify the corresponding reply.

4. PING Output Breakdown

- **TTL (Time to Live):** This is the number of hops (routers) the packet can make before being discarded. Each router along the way decreases the TTL by 1.
- **Round Trip Time (RTT):** The time taken for the packet to go from the source to the destination and back. It's measured in milliseconds.
- **Packet Loss:** This indicates the percentage of packets that were not successfully transmitted or received.

5. Real-World Uses of PING

- **Network troubleshooting:** If you cannot reach a remote server, using `ping` helps identify if the problem is with your device, a router, or the destination host.
- **Latency checking:** If the round-trip time is high, you may have latency issues.

b) Analysis of ICMP and Traceroute Messages

1. Overview of Traceroute

Traceroute is a diagnostic tool used to track the path that packets take from the source to the destination. It sends out ICMP Echo Request messages (or UDP packets in some implementations) and increments the TTL (Time To Live) field in each subsequent message. Each hop along the route (a router or intermediary device) replies with a Time Exceeded message (ICMP Type 11) once the TTL expires.

- **Traceroute Command:** `tracert google.com` or `tracert google.com` (on Windows)

2. Traceroute Process

- **Step 1:** Traceroute starts by sending an ICMP Echo Request with a TTL value of 1 to the destination. The first router along the path receives it, decrements the TTL to 0, and sends an ICMP Time Exceeded message back to the source.
- **Step 2:** The source device receives the Time Exceeded message and increments the TTL to 2 for the next request.
- **Step 3:** The second router does the same and sends a Time Exceeded message back.
- This process continues until the destination host is reached, and the destination sends an ICMP Echo Reply.

The output of a `tracert` (or `tracert` on Windows) might look like this:

Tracing route to google.com [142.250.181.14] over a maximum of 30 hops:

1	<1 ms	<1 ms	<1 ms	192.168.1.1
2	5 ms	4 ms	4 ms	10.10.10.1
3	10 ms	9 ms	9 ms	172.217.9.1
4	12 ms	11 ms	12 ms	142.250.181.14

Trace complete.

3. Breakdown of Traceroute Process

- **Hop 1 (TTL=1):** The first router receives the packet and sends back an **ICMP Time Exceeded** message.
 - ICMP Message Type: 11 (Time Exceeded)
 - Code: 0 (Time to Live exceeded)
- **Hop 2 (TTL=2):** The second router processes the packet and sends back another **ICMP Time Exceeded** message.
- **Final Hop (TTL reaches the destination):** The destination server (e.g., `google.com`) sends an **ICMP Echo Reply** back to the source.

4. Traceroute Output Breakdown

- **IP Address/Hostname:** Each line shows the IP address and sometimes the domain name of the router at each hop.
- **Response Time:** For each hop, the round-trip time (RTT) is shown in milliseconds.
- **Number of Hops:** The number of routers between the source and the destination. It helps to analyze routing paths and detect where network issues might occur.

5. Real-World Uses of Traceroute

- **Network Path Analysis:** Traceroute helps visualize the path data takes across the network and locate possible points of failure or congestion.
- **Network Latency:** By analyzing the RTT at each hop, you can identify which segments of the network are causing delays.
- **Troubleshooting Routing Issues:** If there's an issue with the route taken by packets, traceroute helps pinpoint the problem (e.g., misconfigured routers or network black holes).

Comparison: ICMP PING vs. Traceroute

Feature	PING	Traceroute
Purpose	Test if a host is reachable.	Trace the path packets take to reach a destination.
ICMP Message Type	Echo Request / Echo Reply (Type 8/0).	Time Exceeded (Type 11), Echo Reply (Type 0).
TTL Behavior	Fixed TTL (typically 64 or 128).	TTL starts from 1 and increases with each hop.
Output	Round-trip time (RTT), packet loss.	List of hops along the route with response times.
Use Case	Basic connectivity check.	Detailed analysis of routing path and network issues.

Both tools use ICMP for diagnostics, but **ping** focuses on basic reachability, while **traceroute** gives a deeper analysis of how data travels across the network.